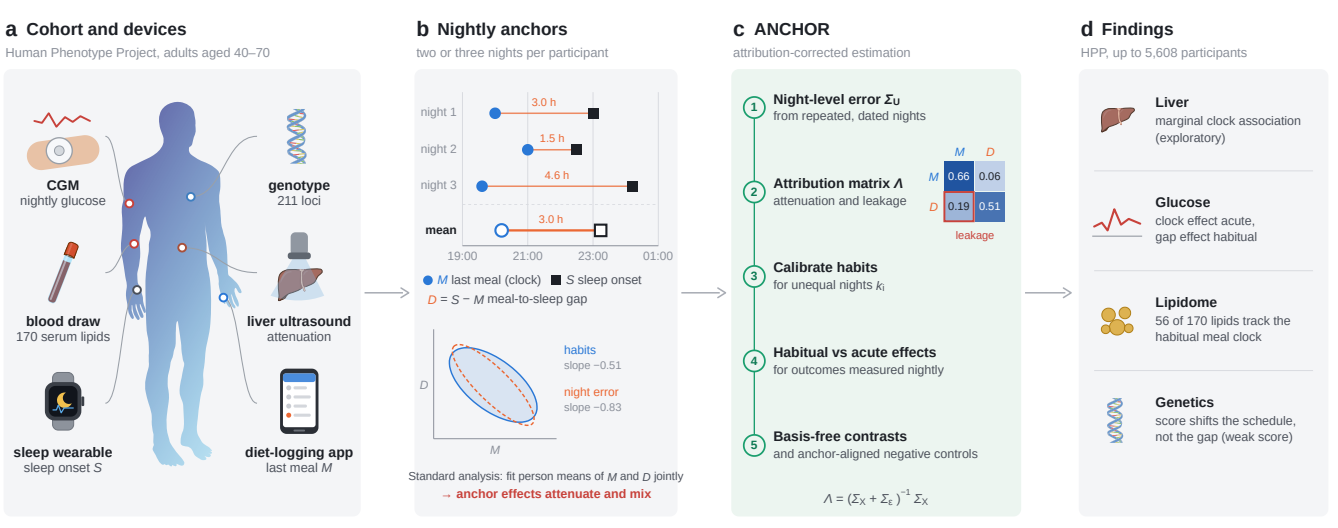

a Cohort and devices
Human Phenotype Project, adults aged 40–70
CGM
nightly glucose
genotype
211 loci
blood draw
170 serum lipids
liver ultrasound
attenuation
sleep wearable
sleep onset S
diet-logging app
last meal M
b Nightly anchors
two or three nights per participant
night 1
night 2
night 3
mean
3.0 h
1.5 h
4.6 h
3.0 h
19:00
21:00
23:00
01:00
M last meal (clock)
S sleep onset
D = S − M meal-to-sleep gap
habits
slope −0.51
night error
slope −0.83
D
M
Standard analysis: fit person means of M and D jointly
→ anchor effects attenuate and mix
c ANCHOR
attribution-corrected estimation
1 Night-level error Σ_U
from repeated, dated nights
2 Attribution matrix Λ
attenuation and leakage
M
D
M
0.66
0.06
D
0.19
0.51
leakage
3 Calibrate habits
for unequal nights k_i
4 Habitual vs acute effects
for outcomes measured nightly
5 Basis-free contrasts
and anchor-aligned negative controls
Λ = (Σ_X + Σ_ε)^−1 Σ_X
d Findings
HPP, up to 5,608 participants
Liver
marginal clock association
(exploratory)
Glucose
clock effect acute,
gap effect habitual
Lipidome
56 of 170 lipids track the
habitual meal clock
Genetics
score shifts the schedule,
not the gap (weak score)